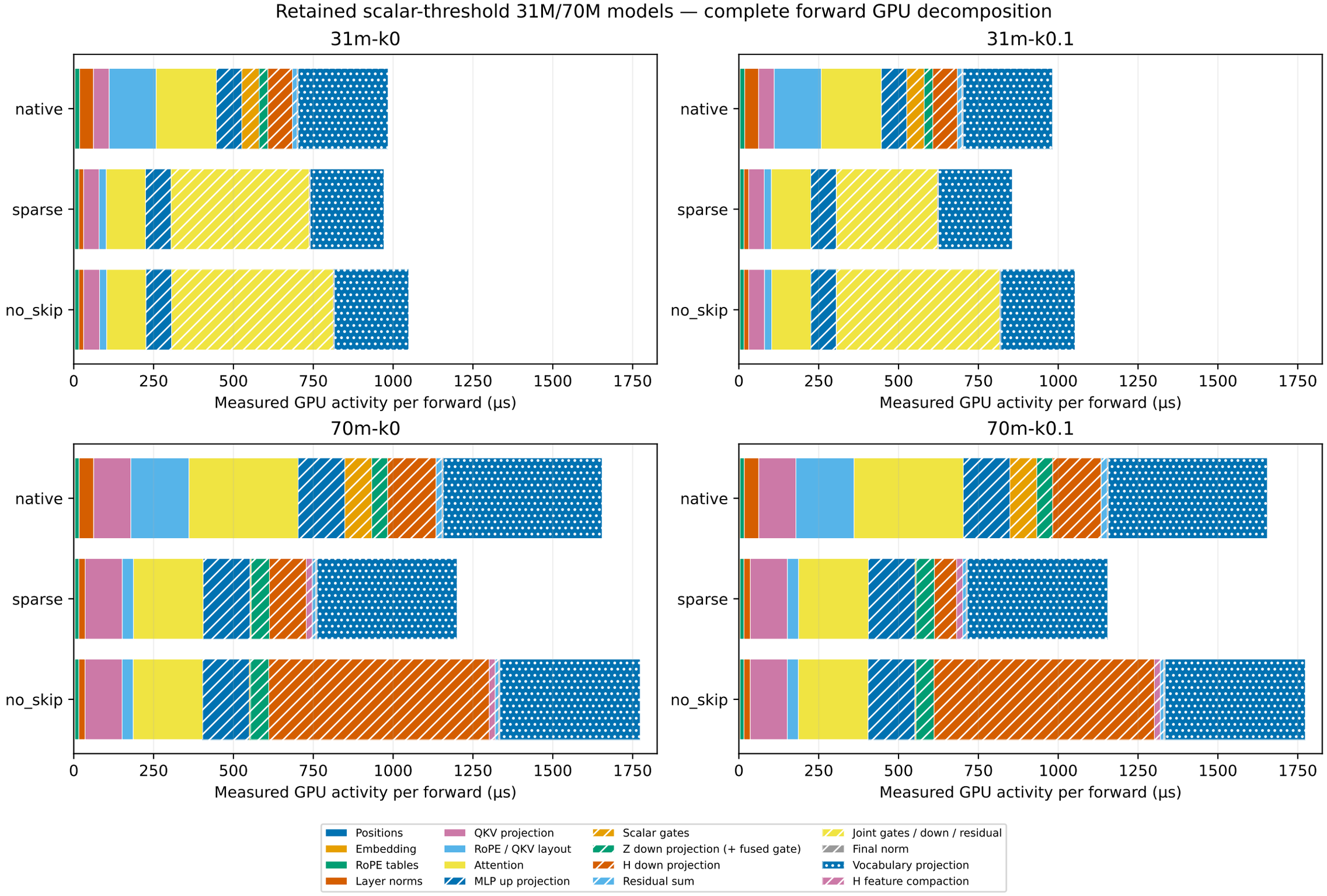
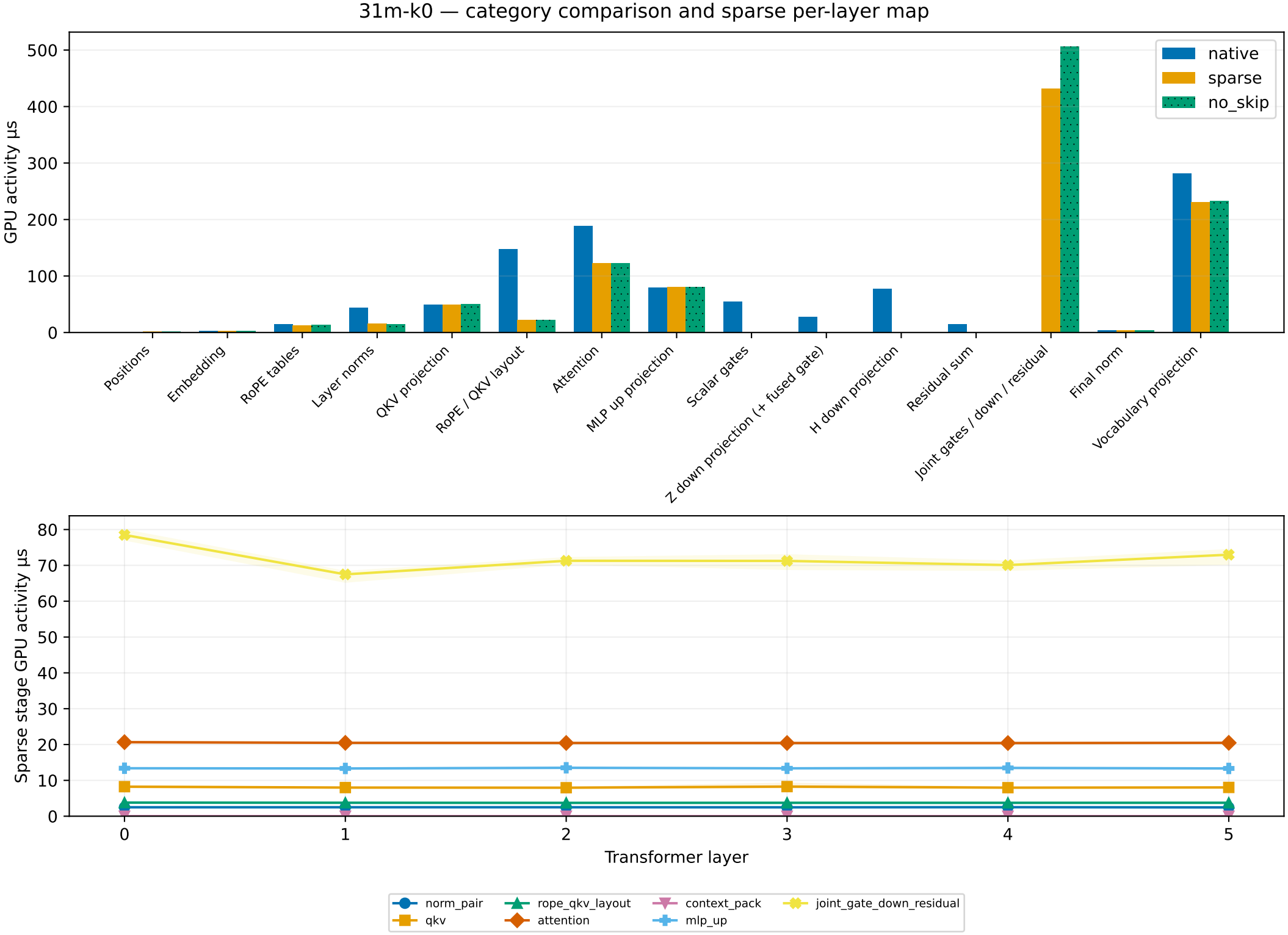
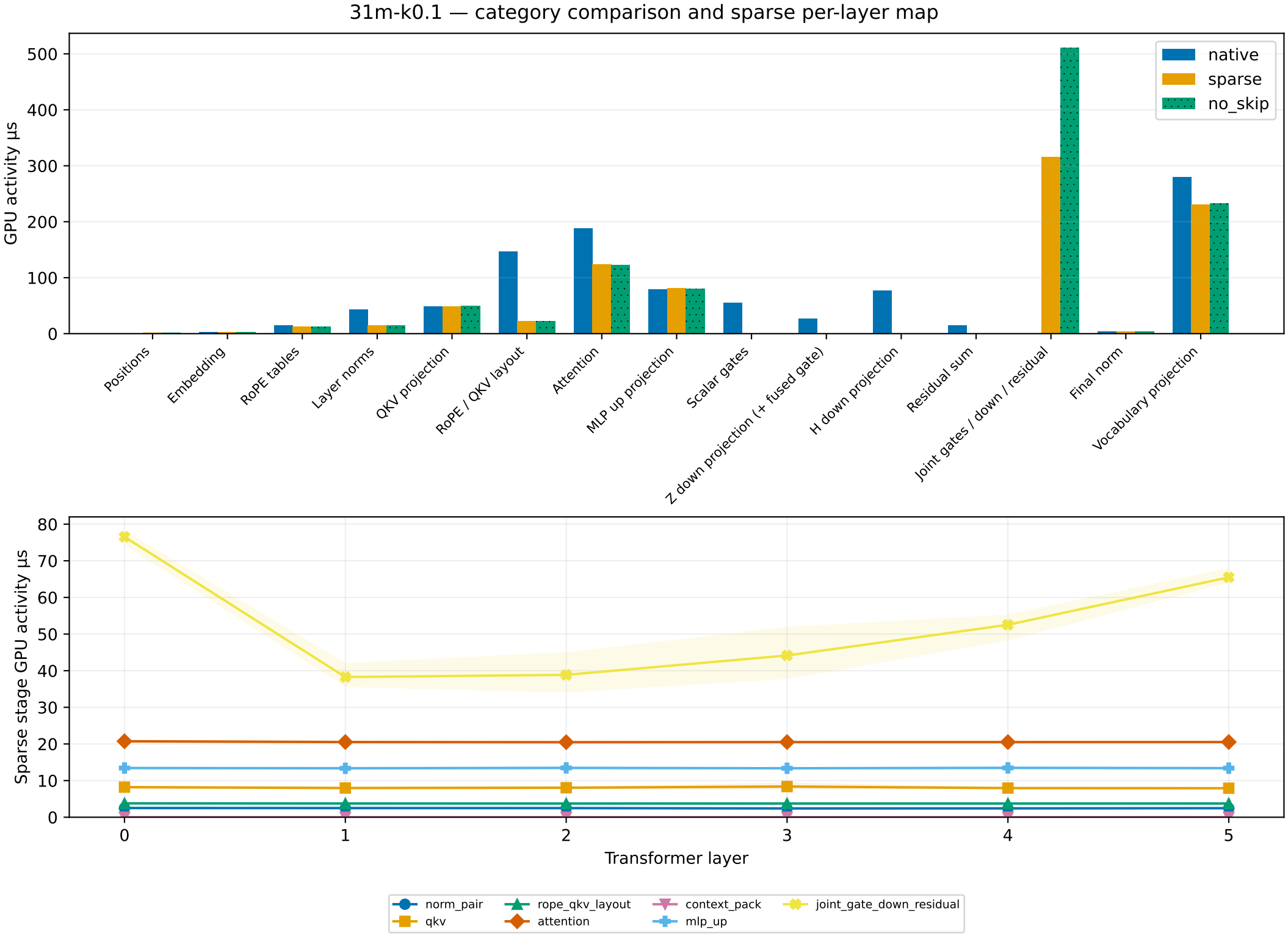
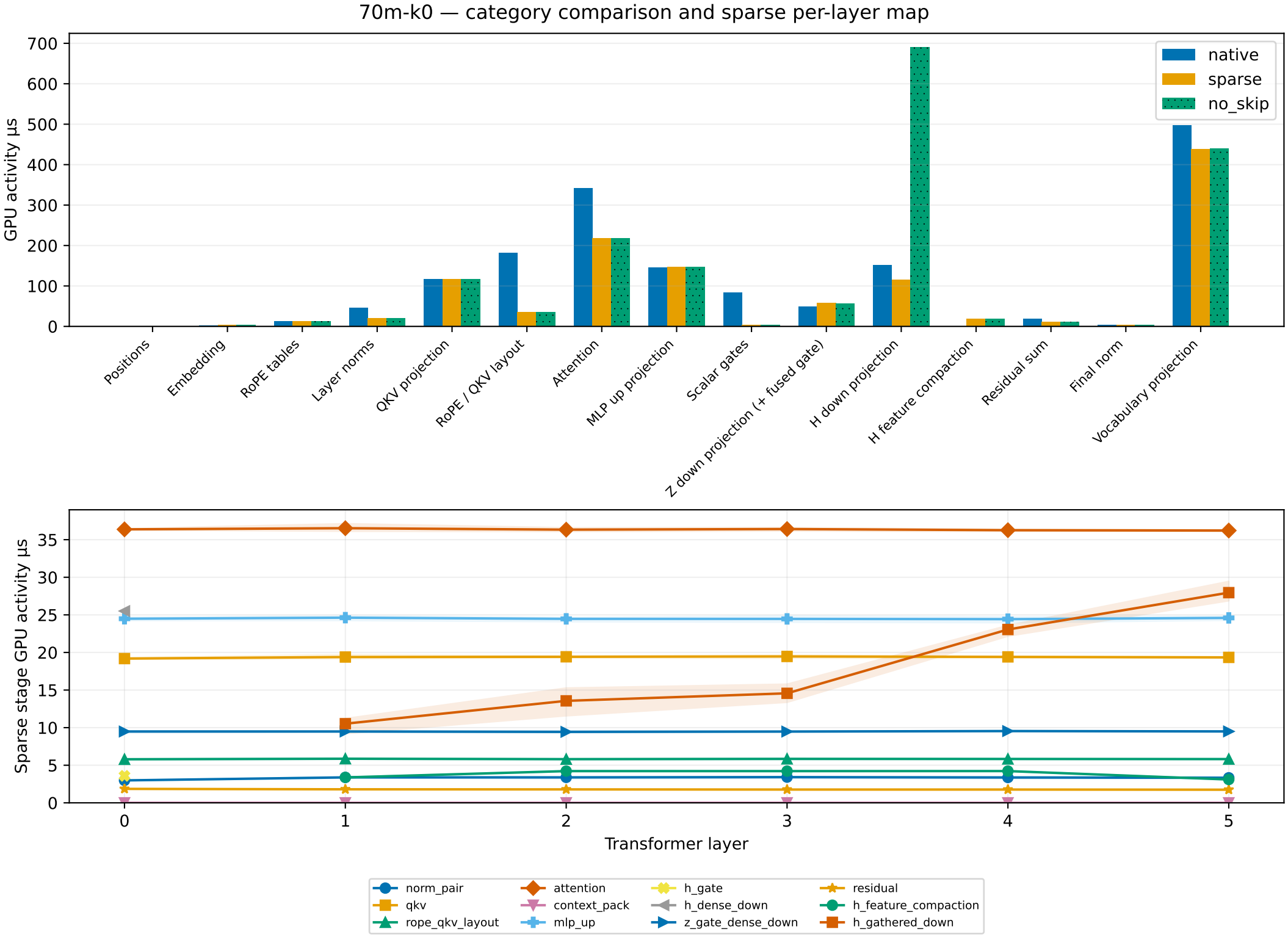
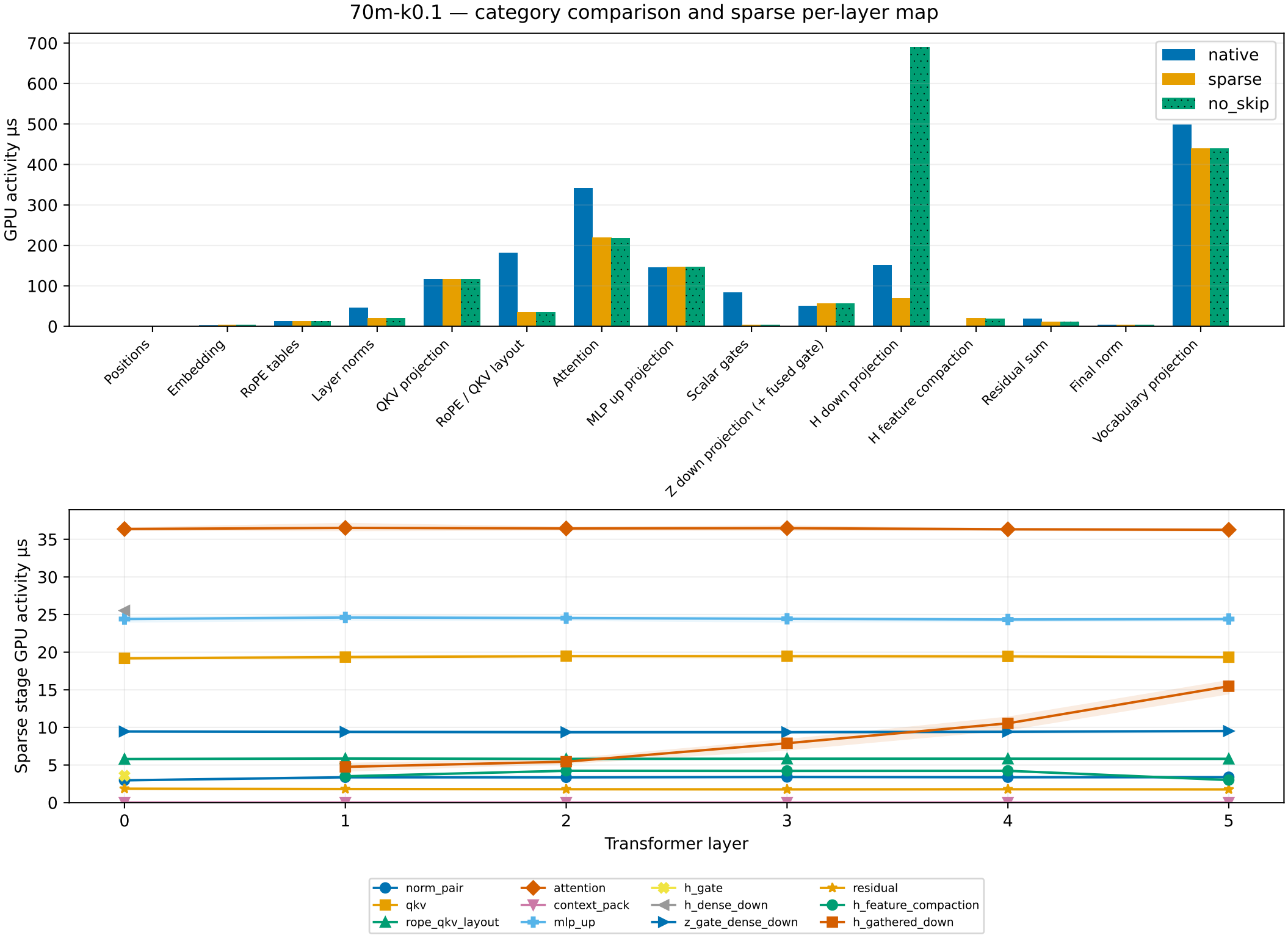

Retained scalar-threshold 31M/70M models — complete forward GPU decomposition
31m-k0
31m-k0.1
70m-k0
70m-k0.1
native
sparse
no_skip
0
250
500
750
1000
1250
1500
1750
Measured GPU activity per forward (µs)
Positions
Embedding
RoPE tables
Layer norms
QKV projection
RoPE / QKV layout
Attention
MLP up projection
Scalar gates
Z down projection (+ fused gate)
H down projection
Residual sum
Joint gates / down / residual
Final norm
Vocabulary projection
H feature compaction


31m-k0 — category comparison and sparse per-layer map
GPU activity µs
native
sparse
no_skip
Positions
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Vocabulary projection
Sparse stage GPU activity µs
Transformer layer
norm_pair
qkv
rope_qkv_layout
attention
context_pack
mlp_up
joint_gate_down_residual


31m-k0.1 — category comparison and sparse per-layer map
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70m-k0 — category comparison and sparse per-layer map
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attention
context_pack
mlp_up
h_gate
h_dense_down
z_gate_dense_down
residual
h_feature_compaction
h_gathered_down


70m-k0.1 — category comparison and sparse per-layer map
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